Python Encapsulation Practice Questions and Answers

Question 1: Student Class

Write a Python program to create a Student class with a private variable __name. Implement set_name() and get_name() methods to set and display the student's name.

Answer:

```
class Student:
  def __init__(self):
    self.__name = ""
  def set_name(self, name):
    self.__name = name
  def get_name(self):
    print("Name:", self.__name)

obj = Student()
obj.set_name("Binty")
obj.get_name()
```

Question 2: BankAccount Class

Write a Python program to create a BankAccount class with a private variable __balance. Implement deposit() and get_balance() methods.

Answer:

```
class BankAccount:
  def __init__(self):
    self.__balance = 0
  def deposit(self, amount):
    self.__balance += amount
  def get_balance(self):
    print("Balance:", self.__balance)

obj = BankAccount()
obj.deposit(5000)
obj.get_balance()
```

Question 3: Employee Class

Write a Python program to create an Employee class with a private variable __salary. Implement methods to set and get salary.

Answer:

```
class Employee:
    def __init__(self):
        self.__salary = 0
    def set_salary(self, salary):
        self.__salary = salary
    def get_salary(self):
        print("Salary:", self.__salary)

obj = Employee()
obj.set_salary(30000)
obj.get_salary()
```

Question 4: Car Class

Write a Python program to create a Car class with a private variable __speed. Implement set_speed() and get_speed() methods.

Answer:

```
class Car:
    def __init__(self):
        self.__speed = 0
    def set_speed(self, speed):
        self.__speed = speed
    def get_speed(self):
        print("Speed:", self.__speed, "km/h")

obj = Car()
obj.set_speed(80)
obj.get_speed()
```

Question 5: PasswordManager Class

Write a Python program to create a PasswordManager class with a private variable __password. Implement methods to set and verify the password.

Answer:

```
class PasswordManager:
    def __init__(self):
        self.__password = ""
    def set_password(self, password):
        self.__password = password
    def verify_password(self, password):
        if self.__password == password:
            print("Password Verified")
        else:
```

```
        print("Incorrect Password")

obj = PasswordManager()
obj.set_password("admin123")
obj.verify_password("admin123")
```

Question 6: Laptop Class

Write a Python program to create a Laptop class with a private variable __brand. Implement set_brand() and get_brand() methods.

Answer:

```
class Laptop:
    def __init__(self):
        self.__brand = ""
    def set_brand(self, brand):
        self.__brand = brand
    def get_brand(self):
        print("Brand:", self.__brand)

obj = Laptop()
obj.set_brand("Dell")
obj.get_brand()
```

Question 7: Book Class

Write a Python program to create a Book class with a private variable __title. Implement set_title() and get_title() methods.

Answer:

```
class Book:
    def __init__(self):
        self.__title = ""
    def set_title(self, title):
        self.__title = title
    def get_title(self):
        print("Title:", self.__title)

obj = Book()
obj.set_title("Python Programming")
obj.get_title()
```

Question 8: Temperature Class

Write a Python program to create a Temperature class with a private variable __celsius. Implement set_temperature() and get_temperature() methods.

Answer:

```
class Temperature:
    def __init__(self):
        self.__celsius = 0
    def set_temperature(self, celsius):
        self.__celsius = celsius
    def get_temperature(self):
        print("Temperature:", self.__celsius, "°C")

obj = Temperature()
obj.set_temperature(25)
obj.get_temperature()
```

Question 9: Movie Class

Write a Python program to create a Movie class with a private variable __rating. Implement set_rating() and get_rating() methods.

Answer:

```
class Movie:
    def __init__(self):
        self.__rating = 0
    def set_rating(self, rating):
        self.__rating = rating
    def get_rating(self):
        print("Rating:", self.__rating)

obj = Movie()
obj.set_rating(4.5)
obj.get_rating()
```

Question 10: Patient Class

Write a Python program to create a Patient class with a private variable __patient_id. Implement set_patient_id() and get_patient_id() methods.

Answer:

```
class Patient:
    def __init__(self):
        self.__patient_id = ""
    def set_patient_id(self, patient_id):
        self.__patient_id = patient_id
    def get_patient_id(self):
        print("Patient ID:", self.__patient_id)
```

```
obj = Patient()
obj.set_patient_id("P101")
obj.get_patient_id()
```